Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
2 P-only DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
3 ANN10 + P-only DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
4 T/P DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
5 ANN10 + T/P DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
6 T/P/ϕ DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
6 T/P/ϕ DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: . Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda1/100/1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 0 complete ICs for all stages within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda / stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**1 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
7 ANN10 + T/P/ϕ DeepONet10	**1 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**100 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**1000 / long**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
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